SIMPLE CONNECTIVITY OF THE LIDMAN–PICCIRILLO PIECE: THE COMPUTATION, STEP BY STEP

BERND J. WUEBBEN

ABSTRACT. This is an expository companion to the computation paper [C] (*Simple connectivity of the Lidman–Piccirillo piece: a certificate-complete computation*). It re-derives, at a much finer grain than a journal format permits, every input of the presentation of $\pi_1(V')$ decided there: the model and its conventions, all three corrected monodromy relations worked end to end, the meridian basings, all four Lagrangian-framing direction words — including the pushoff-basing correction and the monodromy-drift word — the completeness relation, and the assembled presentation at the Lidman–Piccirillo cell. It closes with the Baldridge–Kirk T^4 configuration re-derived by the same method as a fully worked example against published ground truth, and with a protocol for diffing an independently derived presentation against ours. Nothing here is new relative to [C]; the point is that every step be checkable in writing, with no gap a reader must bridge alone.

CONTENTS

1. HOW TO READ THIS DOCUMENT

The computation paper [C] optimizes for auditability: certificates, sweeps, verification pointers. This document optimizes for *followability*: it takes one route through the whole derivation at tutorial speed, in the order in which the objects are actually needed, and works every corrected relation and every direction word explicitly. Each claim of a crossing count or a derived word carries a pointer to the certificate or engine record in [C]'s ancillary files (all public, together with every harness and run log, at https://github.com/bwuebben/exotic-s2xs2); the reader who wants to distrust a step can follow the pointer, and the reader who only wants to *understand* the step should not need to.

Date: July 15, 2026.

Throughout, [C] is the computation paper (July 15, 2026 version — section numbers refer to that version), [R] the reduction paper, and [BK] Baldridge and Kirk, *Constructions of small symplectic 4-manifolds using Luttinger surgery*, J. Differential Geom. 82 (2009) 317–361, arXiv:math/0703065.

2. Conventions, all of them

Convention 2.1.

(1) **Composition.** Paths and loops compose left to right: $\gamma_1\gamma_2$ traverses γ_1 first. A relation written $ugu^{-1} = w$ names the relator $ugu^{-1}w^{-1}$.
(2) **Fiber generators.** x, y, r, s are the edge loops of the rotation-symmetric octagon model of the genus-2 fiber F, based at the vertex point p, with the surface relator $[x, y][r, s]$. In Lidman–Piccirillo's curve names, y realizes the class of b and s the class of d.
(3) **Monodromies.** $h := T_a \circ T_b \colon x \mapsto y^{-1},\ y \mapsto yx$ is the trefoil monodromy on the left handle (machine-certified: it fixes $[x, y]$ exactly, h^6 is the boundary twist, $h^3 = -\mathrm{id}$ on H_1; [C] §2.1). The bundle monodromies are $\tilde{\psi} := h * \mathrm{id}$ (so ψ_0 fixes the *right* handle — in particular the curves e and s — pointwise) and $\tilde{\varphi} :=$ the handle swap ($x \leftrightarrow r$, $y \leftrightarrow s$).
(4) **Base generators.** $\bar{\alpha}, \bar{\beta}$ are the based base loops of Section 4; the presentation's generators are $A := [\bar{\alpha}]^{-1}$ and $B := [\bar{\beta}]^{-1}$. With the left-to-right composition rule this makes conjugation by B implement one *upward* transport around $\bar{\beta}$:

$$BgB^{-1} = \tilde{\psi}(g) \cdot [\text{correction}] \qquad \text{for a fiber loop } g,$$

and likewise A with $\tilde{\varphi}$. The opposite choices are the ε-flips of [R] Remark 2.7; they are swept, but every displayed word below uses the stated convention.
(5) **Swept signs.** Five signs are treated as convention parameters and swept over $\{\pm 1\}^5$ in every decision run: $\varepsilon_3, \varepsilon_4, \varepsilon_5$ (the meridian signs of the three corrected relations) and $\varepsilon_A, \varepsilon_B$ (the filling exponents). The verdict $\pi_1 = 1$ holds for every assignment; no statement below depends on resolving them.
(6) **Meridians.** M and N denote the meridians of the surgery tori T_α and T_β, *based along specific whiskers* fixed in Section 6; "meridian" always means this based loop, not a free-homotopy class.

3. The fiber octagon and the developing engine

Every curve and path in the fiber is specified by a *certificate*: the corner wedge V_i it departs from, the ordered list of octagon edges it crosses, and the arrival wedge V_j. A small engine (`develop.py`, ancillary) converts a certificate into a based word by developing it in the universal cover and reducing with Dehn's algorithm; [C] §3 specifies the five-line algorithm exactly, and its validations V1–V4 pin it against known answers (edge-parallels return their generators; the vertex-link loop develops to 1; the pushoff certificates return $a \simeq x^{-1}$, $b \simeq y$, $e \simeq r^{-1}$, $d \simeq s$ swap-equivariantly; the two lasso arcs differ by a whiskered copy of c).

Three engine outputs are used constantly below, so we fix them now (run record `develop_out.txt`, blocks D3–D5):

- the invariant curve c has based word $(rx)^{-1}$ (at its V_2-whisker), and the curve e develops to r^{-1} (at V_7);

- *position table, s-edge*: traversing s from p, the crossing with c (call it $P_{s'}$) comes *before* the crossing with e (call it s_e). This ordering is forced by disjointness: $c \cap e = \varnothing$, and e's corner arc meets the s-edge near its end. Write s_1 for the initial segment of s up to $P_{s'}$, and s_2 for the initial segment up to s_e; thus $s_1 \subset s_2$.
- *position table, y-edge*: y crosses c once, at c_y; write y_1 for the initial segment of y up to c_y. The loops x and r cross neither c nor e.

4. The cut-square model

The base T_0 (once-punctured torus) is the square $[0,1]^2$ in coordinates (u_1, u_2), minus a puncture disk O at $(3/4, 3/4)$, with the two edge identifications

$$(\xi, (t,1)) \sim (\psi_0 \xi, (t,0)), \qquad (\xi, (1,u)) \sim (\varphi_0 \xi, (0,u))$$

on the total space $F \times [0,1]^2/\sim$. In words: the horizontal edge pair $\{u_2 = 0 \equiv 1\}$ is the *α-cut*, and crossing it *upward* applies ψ_0 to the fiber; the vertical edge pair $\{u_1 = 0 \equiv 1\}$ is the *β-cut*, and crossing it *rightward* applies φ_0.

The basepoint is (p, q) with $q = (1/4, 1/4)$. The based base loops are $\bar\beta =$ the vertical circle through q (travel upward from q, wrap at $u_2 = 1 \equiv 0$, return to q; it crosses the α-cut exactly once and the β-cut never) and $\bar\alpha =$ the horizontal circle through q (crosses the β-cut exactly once, the α-cut never). These realize the minimal possible crossing pattern.

The surgery tori sit *over the cuts*:

$$T_\alpha = c \times \{\alpha\text{-cut circle}\}, \qquad T_\beta = e \times \{\beta\text{-cut circle}\}.$$

Why these are well-defined tori: traveling once along the α-cut circle (a horizontal circle) crosses the β-cut once, applying φ_0, and $\varphi_0(c) = c$ (setwise — the restriction $\varphi_0|_c$ is a free half-rotation of the circle c, a fact that returns in Section 7); traveling once along the β-cut circle crosses the α-cut once, applying ψ_0, and $\psi_0(e) = e$ *pointwise*, so T_β even inherits the product framing. Note the consequence used constantly below:

> *a loop or membrane can meet T_α only over the α-cut, and only at fiber points on c; it can meet T_β only over the β-cut, and only at fiber points on e.*

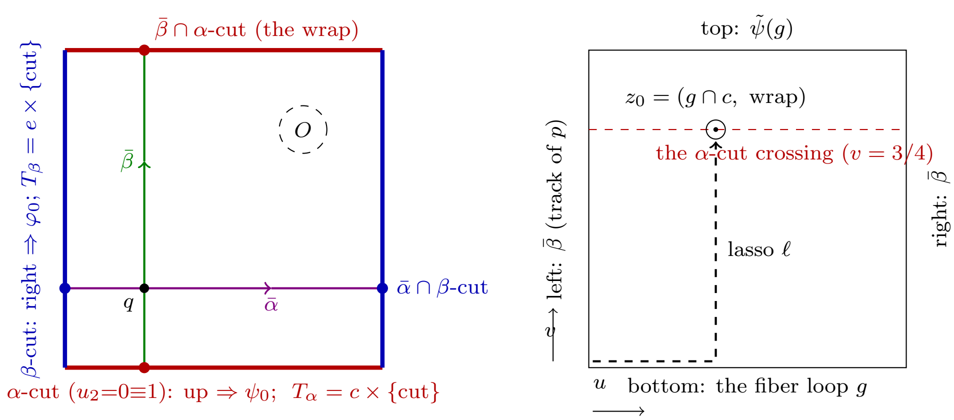

Figure 1. *Illustration only; every crossing count in the text is a position-table or certificate fact.* Left: the cut square. Right: the membrane square of Section 5.

5. Membranes: the mechanism, slowly

Fix a fiber loop g based at p, and transport it once around $\bar\beta$. This sweeps the map of a square

$$\mathfrak{M}_g(u,v) = \big(\text{transport of } g(u) \text{ along } \bar\beta \text{ to time } v\big), \qquad (u,v) \in [0,1]^2,$$

whose four sides are: bottom $= g$; top $= \tilde\psi(g)$ (the loop after one full transport); left and right $= \bar\beta$ itself, the track of the basepoint p (the monodromies fix p, so that track closes). Reading the boundary of the square from its base corner gives the identity

$$g \cdot \bar\beta \cdot \tilde\psi(g)^{-1} \cdot \bar\beta^{-1} = 1 \quad \text{in } \pi_1(R),$$

which after substituting $B = [\bar\beta]^{-1}$ is exactly the monodromy relation $BgB^{-1} = \tilde\psi(g)$. *The square is the proof of the relation*: everything that happens to the relation after surgery happens to this square.

Now remove the surgery tori. If the square's image is disjoint from $T_\alpha \cup T_\beta$, the relation survives verbatim in the complement. If the square meets a torus transversally in interior points $z_1, \ldots, z_k$, puncture it there: the punctured square exhibits the boundary word as a product of small loops around the punctures, each conjugated by an in-square path from the base corner. Each small loop maps to a meridian of the torus it surrounds. Hence in the complement

$$BgB^{-1} = \Big(\prod_i \ell_i\, \mu_i^{\pm 1} \ell_i^{-1}\Big) \cdot \tilde\psi(g),$$

one conjugated meridian per crossing, where ℓ_i is the in-square path to the i-th puncture. Two facts keep this bookkeeping tame:

(1) *The domain is a disk*, so the choice of in-square route to a puncture changes $\ell_i \mu^{\pm 1} \ell_i^{-1}$ by at most powers of the same meridian — i.e., not at all. (This is why the membrane calculus is stable; the only genuinely non-obvious lasso appears when a meridian must be *re-based* to a different whisker, Section 6.)

(2) *The crossings are computable.* By the boxed principle of Section 4, the square $\mathfrak{M}_g$ meets T_α only where its base track crosses the α-cut — for $\bar\beta$, at the single wrap time $v = 3/4$ — and its fiber point lies on c: i.e., at the points $(g \cap c) \times \{\text{wrap}\}$. Likewise for T_β with the β-cut and e. The crossing count of any membrane is therefore (number of base-track crossings with the cut) $\times$ (number of fiber crossings of g with the torus' fiber curve) — both entries read off the position table.

6. The three corrected relations, worked end to end

There are eight monodromy relations: x, y, r, s over $\bar\alpha$ and over $\bar\beta$. We now compute all eight crossing counts and derive the corrected forms. The outcome, stated first:

$$AsA^{-1} = N^{\varepsilon_3}\, y, \qquad ByB^{-1} = M^{\varepsilon_4}\,(yx), \qquad BsB^{-1} = \big(\delta M^{\varepsilon_5} \delta^{-1}\big)\, s \ \text{ with } \ \delta = r^{-1},$$

and the other five relations are clean.

6.1. y **over** $\bar\beta$ **(the template).** Base track: $\bar\beta$ crosses the α-cut once (the wrap), the β-cut never $\Rightarrow$ only T_α is possible. Fiber crossings: $y \cap c = \{c_y\}$, one point. So $\mathfrak{M}_y$ meets T_α in the single transverse point $z = (c_y, \text{wrap})$ and meets T_β not at all. Puncture there. The in-square lasso from the base corner is: along the bottom to $u(c_y)$ — that is the partial loop

y_1 — then vertically up to the wrap. Now *define* the based meridian of T_α along that very whisker:

$$M := y_1 \cdot (\text{meridian circle of } T_\alpha \text{ at } c_y) \cdot y_1^{-1}.$$

With this basing the conjugated-meridian factor *is* $M^{\pm 1}$ — the lasso has been absorbed into the definition — and since $\tilde{\psi}(y) = h(y) = yx$:

$$ByB^{-1} = M^{\varepsilon_4}\,(yx).$$

The sign ε_4 encodes the membrane-orientation convention; it is swept.

6.2. s **over** $\bar{\beta}$ **(the one genuine lasso).** Same base track, so again only T_α, at the wrap. Fiber crossings: $s \cap c = \{P_{s'}\}$, one point. So one puncture, at $z_0 = (P_{s'}, \text{wrap})$, and $\tilde{\psi}(s) = s$ (ψ_0 fixes the right handle pointwise). The in-square lasso is $s_1 \cdot (\text{vertical transport})$. But M is based at c_y, not at $P_{s'}$: to express the correction in terms of M, the meridian must be slid *along the torus* from the crossing to M's basing point — concretely, along the curve c from $P_{s'}$ to c_y. The slide path, developed by the engine, is

$$\delta = s_1 \cdot (\text{arc of } c \text{ from } P_{s'} \text{ to } c_y) \cdot y_1^{-1} = r^{-1}$$

(engine record D3). Hence

$$BsB^{-1} = \left(\delta M^{\varepsilon_5} \delta^{-1}\right) s, \qquad \delta = r^{-1}.$$

Two sanity points. First, c has two arcs from $P_{s'}$ to c_y; the other gives $\delta' = x$, and the engine certifies the consistency identity (V4) $\delta'\delta^{-1} = xr \sim$ a whiskered copy of c — precisely the "slid all the way around" discrepancy one should find. In the group the choice is immaterial: conjugating M by a whiskered c is conjugation by a pushoff class on the boundary 3-torus $\partial\nu(T_\alpha)$, which commutes with the meridian there. The harness fixes $\delta = r^{-1}$. Second, the placement of the correction (left of $\tilde{\psi}(g)$) is an orientation convention; [C]'s 256-case placement sweep shows the verdict does not depend on it.

6.3. s **over** $\bar{\alpha}$**.** Base track: $\bar{\alpha}$ crosses the β-cut once, the α-cut never $\Rightarrow$ only T_β is possible. Fiber crossings: $s \cap e = \{s_e\}$, one point. One puncture; $\tilde{\varphi}(s) = y$. The in-square lasso is $s_2 \cdot (\text{horizontal transport})$, and defining

$$N := s_2 \cdot (\text{meridian circle of } T_\beta \text{ at } s_e) \cdot s_2^{-1}$$

absorbs it, exactly as in §6.1:

$$AsA^{-1} = N^{\varepsilon_3}\,y.$$

6.4. **The five clean relations.** Over $\bar{\beta}$, only T_α is reachable and the fiber condition is "crosses c": x and r do not, so $BxB^{-1} = y^{-1}$ and $BrB^{-1} = r$ hold verbatim. Over $\bar{\alpha}$, only T_β is reachable and the condition is "crosses e": x, y, r do not, so $AxA^{-1} = r$, $AyA^{-1} = s$, $ArA^{-1} = x$ hold verbatim. That is the complete count: three corrections, each at one crossing, five clean relations — and every entry of the count is a position-table fact, not a picture-reading.

7. Meridians, fillings, and the four direction words

A $(1/k)$-Luttinger filling along a curve γ on a surgery torus T kills the class $\mu \cdot (\text{pushoff of } \gamma)^k$, where — this is [BK]'s explicitly stated discipline, p. 3 — *the meridian μ and the Lagrangian-framed pushoff must be based through the same whisker.* Our fillings ($k = \pm 1$ swept as $\varepsilon_A, \varepsilon_B$; directions $\alpha' c^n$ on T_α and $\beta' e^m$ on T_β, $m, n \in \{-1, 0, 1\}$) therefore need four *direction words*: the based pushoffs of the fiber and base directions of each torus, at the same whiskers that base M and N. The four derivations, in increasing order of subtlety:

7.1. **Fiber direction of T_β.** The pushoff of e at the s_2-basing. The engine develops e (departing V_7) to r^{-1}; re-whiskering to the s_2-basing conjugates by s:

$$\mathrm{dir}^{\mathrm{fib}}_{T_\beta} = s\, r^{-1} s^{-1}.$$

(The in-fiber annulus pushoff is the Lagrangian framing here; no base motion, hence no crossing issues.)

7.2. **Fiber direction of T_α.** The pushoff of c at the y_1-basing: the engine's derived word for c,

$$\mathrm{dir}^{\mathrm{fib}}_{T_\alpha} = (rx)^{-1}.$$

7.3. **Base direction of T_α (the drift word).** This is the input with no analogue in any trivial-monodromy calibration. It has since acquired its own known-answer instrument — a Seifert calibration on $\mathrm{MT}(\varphi_0) \times S^1$, surgered along the drift section itself, against classified graph-manifold ground truths ([C], *External calibration of the drift word*; §9 item 1) — which validates the drift machinery externally and shows that the arc pairing discussed at the end of this subsection is basing-dependent, hence a convention. The base-direction curve of T_α travels once along the α-cut circle. But T_α's natural parametrization is not a product: going once around the cut circle applies $\varphi_0|_c$, a *free half-rotation* of c. A closed curve on T_α in the base direction must therefore *drift along half of c* to close up. Following the based-pushoff formula with whisker $\sigma = y_1 c_1$: the transported whisker is $\tilde{\varphi}(y_1)$, and the two identities

$$\varphi_0(c_y) = c_s, \qquad \tilde{\varphi}(y_1) = s_1$$

(the swap carries the y-edge data to the s-edge data) make the fiber part of the closed-up curve *literally the lasso* $\delta = s_1 \cdot (\text{half of } c) \cdot y_1^{-1}$ of §6.2:

$$\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = \lambda = A\,\delta = A\, r^{-1}.$$

Check 7.1 (The drift identity)**.** Squaring must give the full rotation: $\lambda^2 = Ar^{-1}Ar^{-1}$, which the monodromy relations rewrite to $x^{-1}A^2r^{-1}$; abelianized, $2\lambda = 2\alpha' + [c]$ with $[c] = -[r] - [x]$. This is exactly the class a double traverse of the base direction of a half-rotation mapping torus must have — the drift sums to one full copy of c. It holds on the nose, and it is the sharpest internal check available on this word.

The other arc choice gives $Ax = Ar^{-1}(rx)$, a c-multiple — absorbed by the n-sweep of the direction family, so nothing rests on the arc here either. (The Seifert calibration exhibits both pairings concretely: at its own, differently based filling slot the coherent word is the Ax-representative, and the verbatim transplant of Ar^{-1} fails — the arc goes with the basing, not with the curve.)

7.4. **Base direction of T_β (the pushoff-basing correction).** The base-direction curve of T_β is the β-cut circle at fiber s_e, pushed off; its whisker (shared with N) is s_2. Two homotopies are needed to express the based pushoff in the generators, and they have different crossing behavior — this distinction is the content of the correction, and the place where the first version of [C] erred.

Homotopy 1 (the column slide) — clean. Slide the pushoff circle from the cut column to $\bar\beta$'s column $u_1 = 1/4$, at the constant fiber point s_e. The swept annulus lies at fiber s_e throughout; it could meet T_α only if $s_e \in c$, and $c \cap e = \varnothing$; it stays off the β-cut, so it misses T_β too. Clean. (This is the step the original justification "avoids T_α since $c \cap e = \varnothing$" correctly covered — but it is not the only step.)

Homotopy 2 (the whisker transport) — one crossing. What remains is the loop $L = s_2 \cdot V_{s_e} \cdot s_2^{-1}$, where V_{s_e} is the vertical circle at fiber s_e. To express it via B, transport the whisker s_2 once around $\bar\beta$: the membrane is $\mathfrak{A} = \mathfrak{M}_{s_2}$ — *the restriction of the Bs-membrane of §6.2 to $u \le u(s_e)$* — a square with sides s_2, V_{s_e}, $\tilde\psi(s_2) = s_2$ (pointwise), $\bar\beta$. Its crossing count, by the same two-factor principle: it misses T_β (base column $1/4$; interior fiber points $s_2(u)$, $u < u(s_e)$, miss e); it meets T_α at (fiber points of $s_2 \cap c$) $\times$ (the wrap). And here the position table bites: *the c-crossing precedes the e-crossing on the s-edge*, so $P_{s'} \in s_2$, and the membrane meets T_α in exactly the point $z_0 = (P_{s'}, \mathrm{wrap})$ of §6.2 — the same point, with the same local data, because $\mathfrak{A}$ is an oriented restriction of $\mathfrak{M}_s$. The local intersection number is ± 1, so no interior homotopy avoids it.

Puncture and compute, exactly as before. The punctured square gives

$$s_2\, V_{s_e}\, s_2^{-1}\, \bar\beta^{-1} = \ell\, \mu^{\varepsilon_0} \ell^{-1},$$

with ℓ the in-square lasso and ε_0 the same crossing sign that the Bs-membrane assigns to z_0 — i.e., ε_0 is the sweep parameter ε_5. The established Bs-reduction (§6.2) says precisely that $B\, \ell\, \mu^k \ell^{-1} B^{-1} = \delta M^k \delta^{-1}$ for all k. Therefore

$$L = \ell\, \mu^{\varepsilon_5} \ell^{-1} \cdot \bar\beta \quad\Longrightarrow\quad L^{-1} = B \cdot \big(\ell\, \mu^{-\varepsilon_5} \ell^{-1}\big) = B \cdot B^{-1}\big(\delta M^{-\varepsilon_5} \delta^{-1}\big) B = \big(\delta M^{-\varepsilon_5} \delta^{-1}\big)\, B,$$

and since the harness's direction convention has the uncorrected value B (that is, the pushoff circle traversed as $\bar\beta^{-1}$), the honest word is

$$\boxed{\mathrm{dir}^{\mathrm{base}}_{T_\beta} = \big(\delta M^{-\varepsilon_5} \delta^{-1}\big)\, B, \qquad \delta = r^{-1}.}$$

Note what did and did not happen: the B-conjugation cancelled exactly, so the correction sits on the left with the *plain* lasso δ; and the meridian sign came out *anti-coupled* to the Bs-correction sign — a relative statement between two corrections at the same geometric crossing, hence covariant under the global orientation flip that the sign sweep blankets. No new swept convention was introduced. The correction abelianizes to zero, so all homology bookkeeping ([R] Lemma 3.1) is untouched. Every verdict in [C] was re-established with this word; in addition, all eight candidate resolutions (both arcs, both signs, both placements) were swept at the surgery cell before the exact derivation was fixed — all certified trivial — so the theorem never hinged on the resolution, only the derivation's honesty did ([C] §4.5, §7).

8. Completeness in one page, and why the verdict does not need it

The relations of Sections 6–7 are all *true* in $\pi_1(V')$: each is carried by an explicit membrane or filling. A presentation made of true relations presents a group that *surjects* onto $\pi_1(V')$. Since quotients of the trivial group are trivial, **a trivial verdict is valid**

without any completeness argument. Completeness matters only for a nontrivial verdict (a missing relation could inflate the group), and [C] §6 proves it anyway, via a graph-of-groups decomposition over the cut system; the Tietze reduction of that decomposition returns the relations above plus exactly one more,

$$R_3: \quad B\,\kappa_3\,B^{-1} = \tilde{\psi}(\kappa_3), \qquad \kappa_3 = s^{-1}r^{-1}yx, \;\; \tilde{\psi}(\kappa_3) = r^{-1}s^{-1}x,$$

the transport relation of the third basis element of $\pi_1(F \smallsetminus \nu c)$ (engine record D5) — true for the same membrane reasons, and diagram-independent. (Practical note: R_3 also makes the presentations far friendlier to rewriting-system completion; without it Knuth–Bendix fails to reach confluence on several cells — an *inconclusive* outcome, never a wrong verdict.)

9. The assembled presentation at the LP cell

At $(m,n) = (0,0)$, generators x, y, r, s, A, B, M, N and relators:

	relator	derived in
R0	$[x,y][r,s]$	§2
A1–A3	$AxA^{-1} = r,\; AyA^{-1} = s,\; ArA^{-1} = x$	§6.4
B1–B2	$BxB^{-1} = y^{-1},\; BrB^{-1} = r$	§6.4
R3	$B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$	§8
C1	$AsA^{-1} = N^{\varepsilon_3}y$	§6.3
C2	$ByB^{-1} = M^{\varepsilon_4}(yx)$	§6.1
C3	$BsB^{-1} = (r^{-1}M^{\varepsilon_5}r)\,s$	§6.2
F1	$M\,(Ar^{-1})^{\varepsilon_A} = 1$	§7.3
F2	$N\,\big((r^{-1}M^{-\varepsilon_5}r)B\big)^{\varepsilon_B} = 1$	§7.4

For $m, n \neq 0$ the filling words acquire the fiber factors $((rx)^{-1})^n$ resp. $(sr^{-1}s^{-1})^m$. Known-answer probes: dropping F1–F2 gives $H_1 = \mathbb{Z}^2$; dropping one filling gives $H_1 = \mathbb{Z}$; the full presentation has $H_1 = 0$ in every convention — each exactly as [R] Lemma 3.1 requires. Decision: coset enumeration certifies $|G| = 1$ at the LP cell in all 32 sign conventions; Knuth–Bendix reduction certificates (every generator explicitly reduced to the identity) re-derive the whole grid — 288/288, and 576/576 for the independently derived second diagram ([C] §7).

10. The same method on Baldridge–Kirk's T^4: a fully worked calibration

Everything above, minus the monodromy, can be tested against published ground truth. Take [BK] §2: $T^4 = \hat{H} \times \hat{K}$ with fiber $\hat{H}$ ($\pi_1 = \langle x, y \mid [x,y] \rangle$) and base $\hat{K}$ ($\pi_1 = \langle a, b \rangle$, based loops $\bar{a}, \bar{b}$), and the disjoint Lagrangian tori

$$T_1 = X \times A_1, \qquad T_2 = Y \times A_2,$$

X, Y parallel pushoffs of x, y in the fiber, $A_1 = \{u_2 = 0.3\}$, $A_2 = \{u_2 = 0.7\}$ parallel copies of a in the base. Trivial monodromy; the membrane calculus of Sections 5–6 applies verbatim with cuts along a and b.

Crossing counts. The membranes of $\bar{a}$-transport are clean ($\bar{a}$ is parallel to A_1, A_2: no base crossing). For $\bar{b}$-transport: the base track crosses A_1 once and A_2 once; the fiber conditions are $x \cap Y$, $y \cap X$ ($X \cap Y$ is one point). Result: exactly two corrected relations,

$$b\,y\,b^{-1} = \mu_1^{\pm 1}\,y \;\;\text{(from } T_1\text{)}, \qquad b\,x\,b^{-1} = \mu_2^{\pm 1}\,x \;\;\text{(from } T_2\text{)},$$

each meridian based along its in-membrane whisker — the exact analogue of M, N.

The lasso datum. The fillings kill $\mu_i \cdot (\text{direction word})^{\pm 1}$ with meridian and pushoff based through the same whisker. For T_1 the direct basing is clean. For T_2 it is *not*: sliding the fiber-pushoff circle Y' down the base whisker (from $u_2 \approx 0.7$ to the basepoint row) transports the whole circle Y', and Y' contains the point $Y \cap X$; the slide therefore crosses $T_1 = X \times A_1$ once, at $(Y \cap X) \times A_1$. Exactly the Homotopy-2 phenomenon of §7.4: testing the slide on the circle's *basepoint* alone misses it. The repair is one lasso: the T_2 filling uses the b-conjugated meridian (equivalently, [BK]'s own $\ell_2 = bab^{-1}$ records the same wrap). The derived lasso pair is $(d_1, d_2) = (1, b)$; the $T_1 \leftrightarrow T_2$ mirror $(b, 1)$ is equivalent by an exact symmetry of the configuration.

Ground truth and outcome. [BK] Lemma 2: double ± 1 surgery gives $\pi_1 = (\mathbb{Z}^2 \rtimes_A \mathbb{Z}) \times \mathbb{Z}$ with $|\mathrm{tr}\, A| \in \{1, 3\}$ by the relative signs — non-abelian in *every* convention; single surgery gives $H_3(\mathbb{Z}) \times \mathbb{Z}$. Sweeping all 64 sign/placement conventions per lasso pair (harness `decide_t4.g`, run record ancillary):

$(d_1, d_2) = (1, b)$ and its mirror $(b, 1)$:	64/64 match ground truth, with the exact $\mathrm{tr}\,3/\mathrm{tr}\,1$ split;
all seven other pairs :	0/64 — six with *half* their failures abelian; (b^{-1}, b^{-1}) with wrong non-abelian groups throughout.

Single surgery: 8/8; the no-filling probe returns $H_1 = \mathbb{Z}^4$; and [BK]'s own published words, run through the same decision battery, match ours cell by cell. Three lessons: (i) the membrane/basing/filling calculus reproduces a community-vetted computation exactly; (ii) a *single* missed whisker crossing is reliably fatal and reliably *detected* — and abelian output is its signature (this is also why any "can't be abelian" theorem on T^4 is evidence about *wrong words*, not about methods that derive the right ones); (iii) the one class of input this calibration cannot exercise is monodromy drift (§7.3) — T^4 has none — which is why that word received its own known-answer instrument: the Seifert calibration of [C] (surgery along the drift section of $\mathrm{MT}(\varphi_0) \times S^1$, ground truths from the classification of torus twists), which reproduces classified groups with the coherent words and separates seven wrong-word families, including the whisker-conjugate family every earlier instrument left undecided.

11. How to diff a presentation against ours

Suppose an independently derived presentation of $\pi_1(V)$ is at hand. To compare it with Section 9 relator by relator, quotient out the four benign discrepancies first:

1. *global inversions* of any generator (direction conventions; our ε-flips);
2. *the five swept signs* $\varepsilon_3, \varepsilon_4, \varepsilon_5, \varepsilon_A, \varepsilon_B$;
3. *basing conjugation*: any relator may be conjugated, and any meridian re-whiskered ($\mu \mapsto w\mu w^{-1}$), without geometric content;
4. *the arc identity*: $\delta = r^{-1}$ vs. $\delta' = x$ differ by a whiskered copy of c, immaterial in the group (§6.2).

A relator that fails to match modulo (1)–(4) is *the* disagreement, localized to one geometric datum — one crossing count, one lasso, one framing word — and each of ours carries the derivation above plus a certificate pointer in [C]. Conversely, two structural checks apply to any candidate presentation of this manifold: its abelianization must behave as in Section 9's known-answer probes, and — since presentations by true relations surject onto π_1 — a *nontrivial* verdict carries the burden of completeness, while a trivial one does not (§8).

Certificate index. Position tables and derived words: `develop_out.txt` (D3 = lassos, D4 = positions, D5 = κ_3), validations V1–V4 ([C] §3). Decision records: `honest_run_2.log`, `decide2_out2.txt`, `kb_certify_out2.txt`, `kb_diag2_full_out.txt`, `placement_out2.txt`. T^4: `decide_t4.g`, `t4_out.txt`. All at https://github.com/bwuebben/exotic-s2xs2.

Bernd J. Wuebben, New York, NY. wuebben@gmail.com